

Camera fault detection for Autonomous Delivery Vehicles

Background

Autonomous Delivery Vehicles (ADV) are self-driving systems that use onboard computers and sensors to perceive the environment and control the vehicle without continuous human input. Unlike conventional cars driven manually, ADVs follow machine-readable instructions and real-time perception to execute complex driving tasks.



Figure 1. Autonomous Delivery Vehicles.

Autonomous driving systems are typically organized into three core modules: perception, decision-making, and actuation. The perception module fuses data from multiple sensors—such as cameras, LiDAR, and millimeter-wave radar—to build a real-time view of the environment. The decision-making module, running on high-performance computing units with AI algorithms, performs path planning, behavior prediction, and risk assessment. The actuation module then controls the vehicle's steering, throttle, and braking hardware to execute the planned maneuvers. Working together, these components enable safe, efficient navigation in complex traffic scenarios. Thus ADVs rely heavily on a robust sensor suite to perceive the world and make timely decisions.



Figure 2. Camera of ADVs.

Among these sensors, cameras are pivotal, supplying the visual information needed for object detection, lane keeping, and scene understanding. Unlike conventional vehicles, ADVs require sensors to operate continuously and reliably to maintain safety and functionality. However, cameras are susceptible to faults—such as degraded image quality, intermittent signal loss, or complete outages—that can compromise system performance and, in extreme cases, lead to incidents.

Integrating Prognostics and Health Management (PHM) into autonomous driving is therefore essential for maintaining reliability and safety. PHM continuously monitors system health using real-time data analytics and predictive modeling. By identifying potential issues before they manifest as failures, PHM enables proactive maintenance, reduces unplanned downtime, and optimizes resource allocation—capabilities that are especially critical in autonomy, where sensor faults can have immediate and serious consequences.

In the context of ADV cameras, PHM focuses on operational indicators—such as current draw, GPU/CPU utilization, signal quality, and error counters/logs—to forecast emerging problems. These parameters are continuously tracked to detect early signs of degradation or malfunction. For example, unusual current spikes or deteriorating signal metrics may indicate issues that could escalate into critical failures. By analyzing such patterns, a PHM system can issue maintenance alerts, trigger safe fallback or de-rating strategies, prevent costly repairs, and enhance the overall resilience of the autonomous driving stack.

The objective of this project

As a key component of ADVs, the onboard camera plays a pivotal role: it provides the visual perception required for object detection and tracking, lane-marking and road-edge detection, traffic-light/sign recognition, free-space segmentation, and scene understanding that support prediction, planning, and safe control. Under fault conditions, an ADV camera may produce incorrect recognition results, introducing safety risks. As shown in the figure below, a camera under fault condition misclassified cars as road construction plate.

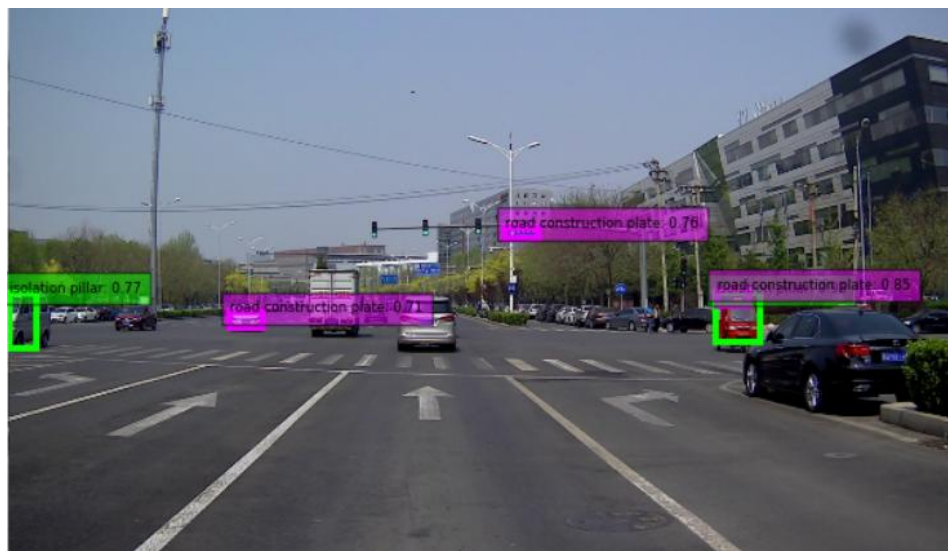


Figure 3. “Car” is wrongly recognized as “road construction plate”.

Accordingly, this project aims to **evaluate the health status of ADV cameras to ensure safe operation.**

For a camera installed on an ADV, its health state—determined from runtime/telemetry parameters—can be categorized as shown in the table below:

- Normal operation

- Warning (early signs of anomaly)
- Degraded performance (partial loss of function/quality)
- Shutdown or unreliable output (stream unavailable or not trustworthy)

• **Table 1. Four health classes of ADV cameras**

Label	Health conditions of ADV cameras
1	Normal operation
2	Warning
3	Degraded performance
4	Shutdown or unreliable output

Data Description

In this study, we collected runtime data from the onboard camera to build the dataset. The data was collected over multiple measurement cycles, with each cycle consisting of 60 measurements (time index 1~60). Within a single measurement cycle, we obtain time-varying data for Current, GPU usage, and CPU usage (time series data), as well as scalar data for EyeQuality, DecodeErrCount, and miniFps (constant). **We determine the fault state based on the data from one measurement cycle, which consists of three time-series vectors and three scalar features.**

• **Table 2. Example sample**

Current	gpuUsage	cpuUsage	EyeQuality	DecodeErrCount	miniFps	Time index	label
90.336	62.4	47	161.8	26	10	1	1
155	85	92	161.8	26	10	2	1
170	40	75	161.8	26	10	3	1
180	40	25	161.8	26	10	4	1
...	...	...	...	...	...	...	...
135	52	60	161.8	26	10	59	1
128	68	78	161.8	26	10	60	1

The feature dimensions are as follows:

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- ***Name (code): current consumption (Current)***
 - *Unit: mA*
 - *Typical range: 50–210*
 - *Meaning: Power draw of the camera/module and link; sustained increases or decreases may indicate power-supply or hardware issues.*
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- ***Name (code): GPU utilization (gpuUsage)***
 - *Unit: %*
 - *Typical range: 10–99*
 - *Meaning: Compute load for perception/codec; persistently near the upper bound may indicate resource pressure or overload.*
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- ***Name (code): CPU utilization (cpuUsage)***
 - *Unit: %*
 - *Typical range: 30–80*
 - *Meaning: General compute load; abnormally high or persistently below expectation may reflect scheduling or performance problems.*
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- ***Name (code): Signal eye-diagram quality (EyeQuality)***
 - *Unit: dimensionless (1)*
 - *Typical range: 120–180*
 - *Meaning: Integrity margin of the video/data link; decreases suggest jitter, noise, crosstalk, or connection issues.*
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- ***Name (code): Decode error count (DecodeErrCount)***
 - *Unit: count (1)*
 - *Typical range: 0–50*
 - *Meaning: Stream errors and retransmissions; increases indicate link degradation or bitstream corruption.*
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- ***Name (code): Main-link frame rate (miniFps)***
 - *Unit: Hz*
 - *Typical range: 10–11*
 - *Meaning: Actual output frame rate of the monitored link; deviations from the typical range may signal faults.*
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Problem definition

This project needs to identify different health classes. Please design classification algorithms to identify health classes of Cameras, while discussing the effect of feature engineering operations. You can utilize any knowledge which you have learned in this course. Of course, you can also utilize or even design new algorithms that is out of the scope of this course.

Students should attempt some methods for fault prediction and conduct comparisons and analyses of these approaches. Moreover, they should also discuss how different feature engineering operations impact the model's performance.

Group

All the students are expected in groups of 2 or 3 students. The project presentation and Q&A session will be held in the lecture on December 30th (16th week). There is no specific requirement for the algorithm.

Report Submissions

This Final-term Project requires students to submit three reports.

- i. The first report is a proposal, and the specific requirements can be found in the assignment document (Assignment - Proposal Submission.pdf) on the online classroom website.
- ii. The second report is a preliminary report, and its specific requirements will be made available after the proposal.
- iii. The third report is the final report, and its specific requirements will be disclosed after the preliminary report.

Please note that the submission deadlines for each report are different. Students can check the latest course schedule on the online classroom website to find out the submission dates for these reports.

Presentation Submissions

This presentation requires the submission of **all relevant runnable codes (annotate your codes when programming) and the PPT slides for the final presentation.**

Presentation

5 minutes for presentation and 2 minutes for Q&A.

Scoring system

The scoring criteria for this project presentation are as follows:

Scoring indicator	Scores (Out of 100 points)	Score
Proposal	5	
Preliminary Report	15	
Presentation	20	
Personal contribution	10	
Final Report	50	